

ATPL EASA STUDY GUIDE | 010 - AIR LAW

Chapter 14: Air Traffic Services & Emergencies

Syllabus Source: CAE Oxford (pp. 279-296) | Pass Target: >= 90% AviationExam

1. THE THREE CORE AIR TRAFFIC SERVICES (ATS)

- 1. Air Traffic Control (ATC): Prevents collisions between aircraft and between aircraft and obstacles on the maneuvering area, and expedites and maintains an orderly flow of traffic. (Subdivided into: Area Control ACC, Approach Control APP, Aerodrome Control TWR).
- 2. Flight Information Service (FIS): Provides advice and information useful for the safe and efficient conduct of flights (weather, changes in serviceability of nav aids, traffic information).
- 3. Alerting Service (ALRS): Notifies appropriate organizations regarding aircraft in need of search and rescue aid, and assists such organizations as required. Provided to ALL aircraft provided with ATC, or submitting a flight plan.

2. AUTOMATIC TERMINAL INFORMATION SERVICE (ATIS)

Continuous broadcast of routine non-control information at busy aerodromes (Voice-ATIS and D-ATIS): - Identified by a designator letter from the ICAO spelling alphabet (Alpha to Zulu), updated sequentially. - Updated whenever a significant change in meteorological or operational conditions occurs. - Pilot must acknowledge current ATIS code upon initial contact with ATC.

3. THE THREE EMERGENCY PHASES (ANNEX 11 / 12)

Emergency Phase	Official Activation Criteria (AviationExam Core)
1. INCERFA (Uncertainty Phase)	No communication received within 30 minutes after time it should have been received, OR aircraft fails to arrive within 30 minutes of ETA.
2. ALERFA (Alert Phase)	Following INCERFA, attempts to communicate fail; OR aircraft cleared to land fails to land within 5 minutes of ETA; OR operating efficiency impaired.
3. DETRESFA (Distress Phase)	Following ALERFA, or fuel on board considered exhausted, or information indicates aircraft about to make forced landing or has crashed.

4. MANDATORY READBACK REQUIREMENTS (SERA.8015)

The flight crew shall read back to the air traffic controller safety-related parts of ATC clearances and instructions which are transmitted orally. The following parts shall ALWAYS be read back:

- 1. Route Clearances: Any ATC route clearance, SID, STAR, or route modification.
- 2. Clearances and Instructions to Enter, Land On, Take Off On, Hold Short Of, Cross, or Taxi On: Any runway instruction must be read back completely with runway designator.
- 3. Runway-in-use, Altimeter Settings, SSR Codes: Crucial numerical flight parameters.
- 4. Newly Assigned Level Instructions, Heading and Speed Instructions: Vertical and horizontal vectors.
- 5. Transition Levels: Whether given by ATC or ATIS.

5. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: What condition triggers the declaration of INCERFA (Uncertainty Phase)?

[A] Aircraft fails to land within 5 minutes of landing clearance. [B] No communication within 30 minutes after time it should have been received. [C] Fuel on board is below reserve. [D] Pilot squawks 7600.

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. INCERFA is declared when no communication has been received within 30 minutes of the time it should have been received.

Q2: An aircraft that has been cleared to land fails to land within what time interval to trigger ALERFA?

[A] 2 minutes [B] 5 minutes [C] 10 minutes [D] 15 minutes

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. An aircraft cleared to land that fails to land within 5 minutes of estimated landing time triggers an immediate Alert Phase (ALERFA).

Q3: When the fuel on board is considered exhausted, which emergency phase is declared?

[A] INCERFA [B] ALERFA [C] DETRESFA [D] PAN PAN

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [C]. Fuel exhaustion triggers the Distress Phase (DETRESFA).

Q4: To whom is the Alerting Service (ALRS) provided?

[A] Only to IFR flights in Class A airspace [B] To all aircraft provided with air traffic control service and to any other aircraft submitting a flight plan [C] Exclusively to commercial passenger airliners [D] Only during search and rescue operations

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Alerting service is provided to all aircraft receiving ATC service, having filed a flight plan, or otherwise known to ATS.

Q5: Which ATS unit provides air traffic control service to en-route controlled flights?

[A] Aerodrome Control Tower (TWR) [B] Approach Control Unit (APP) [C] Area Control Centre (ACC) [D] Flight Information Centre (FIC)

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [C]. The Area Control Centre (ACC) is responsible for en-route air traffic control.

Q6: What emergency frequency must be continuously monitored by aircraft equipped with VHF radio when operating in Class A airspace?

[A] 121.5 MHz [B] 123.45 MHz [C] 118.0 MHz [D] 136.0 MHz

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [A]. Under SERA and ICAO Annex 10, 121.5 MHz is the international VHF aeronautical emergency frequency and must be monitored whenever practicable.